



Platform Core Presentation

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Our Role As Platform Core

Design Core Systems

Implement the 4 core parts of the program that we outlined in the planning diagrams, Database, Leaderbaord, Game Registry, and User Identity

Keep Other Teams In the Loop

Communicate with other teams on where we are at with the core systems, so that they can be tested, and have UI designed and implemented for them.

Frequently Update Documentation

Update the planning diagram documents with any changes we make to the core systems, so that everyone is on the same page.

Design Critique

What We Found

GitLab template structure differs significantly from project structure

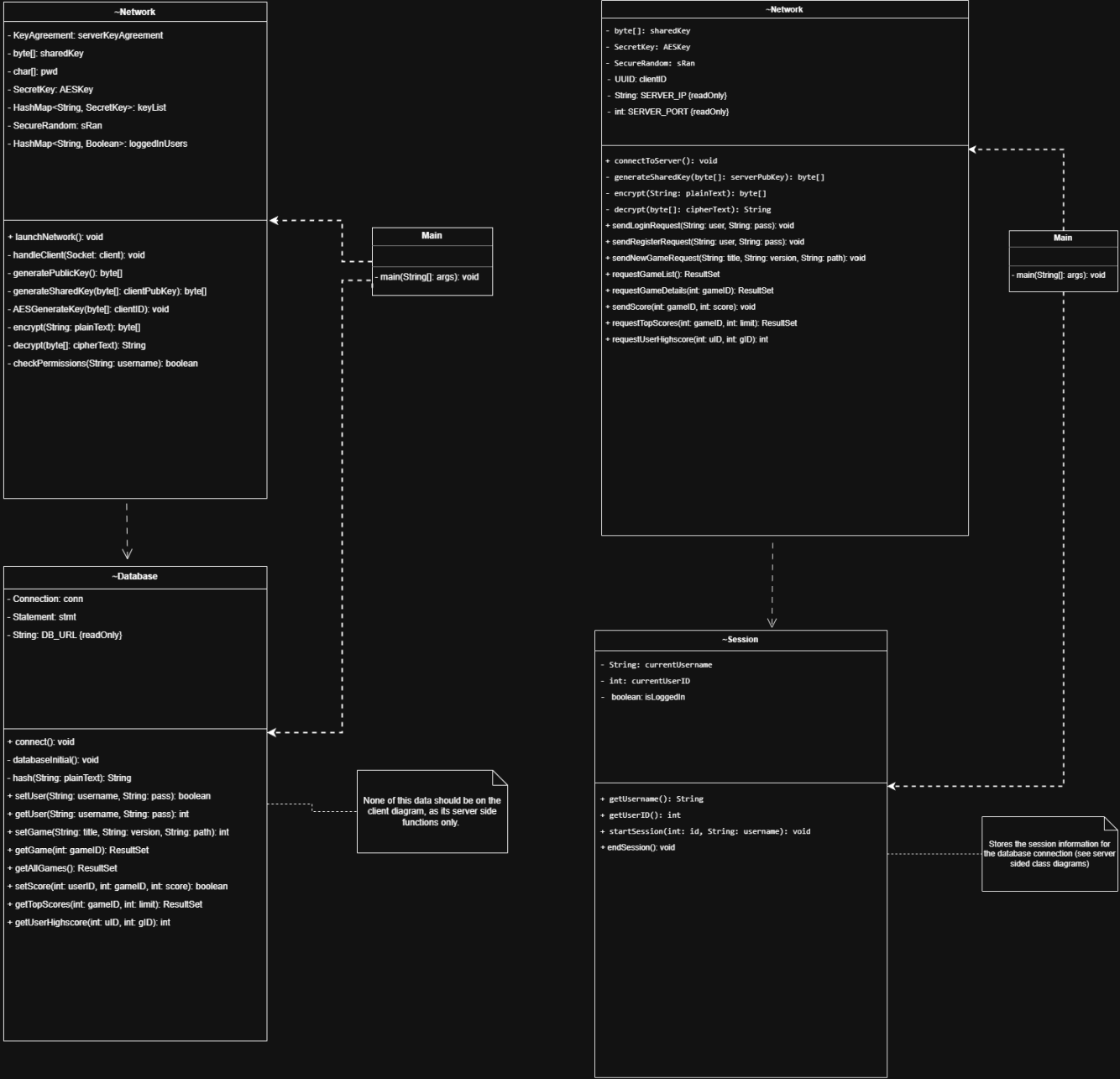
Our diagrams focus on database integration and interactions between client and server.

The setup for this major component of the system was not laid out in the startup diagrams.

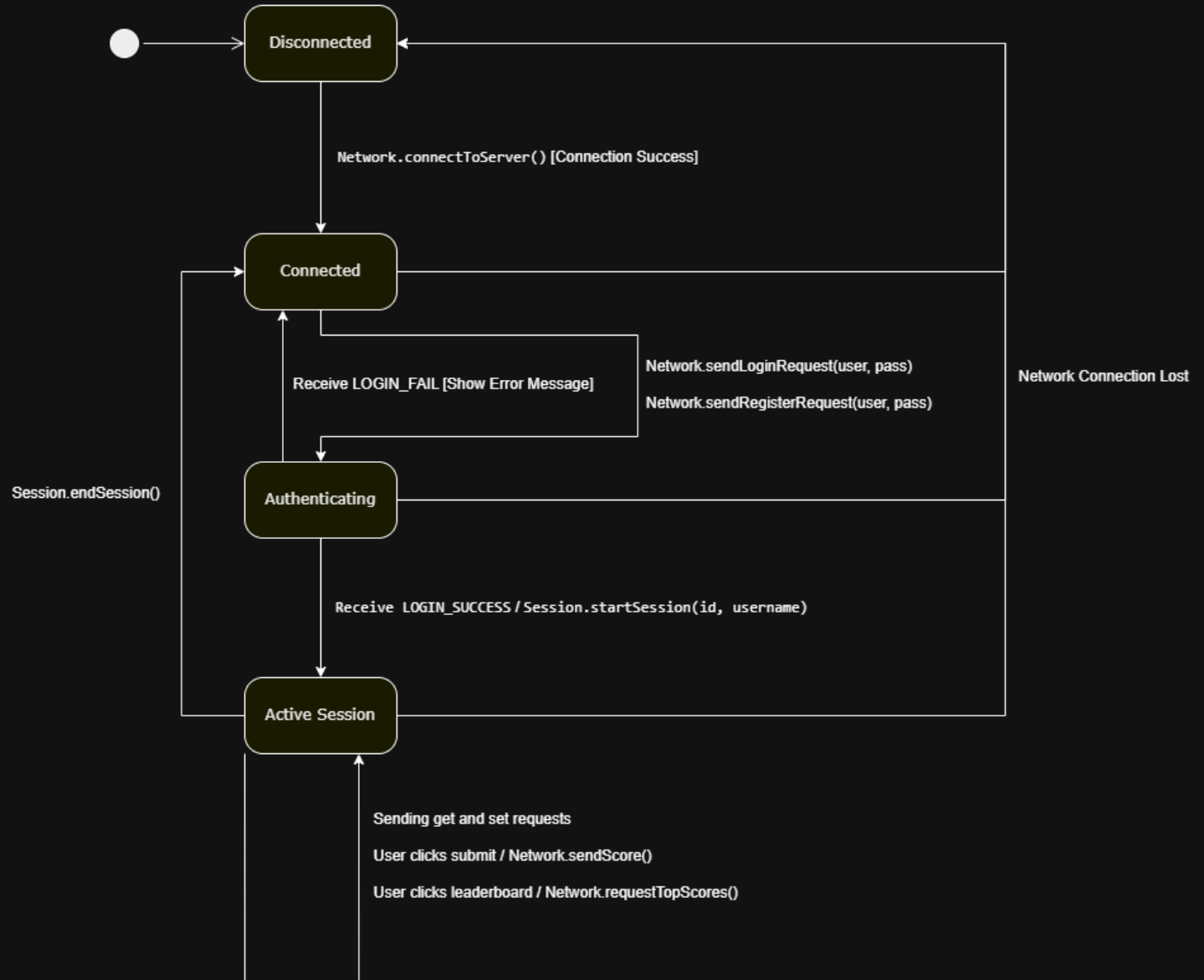
Database From a Glance

- **Server vs client**
- **Dependencies**
- **Overview**

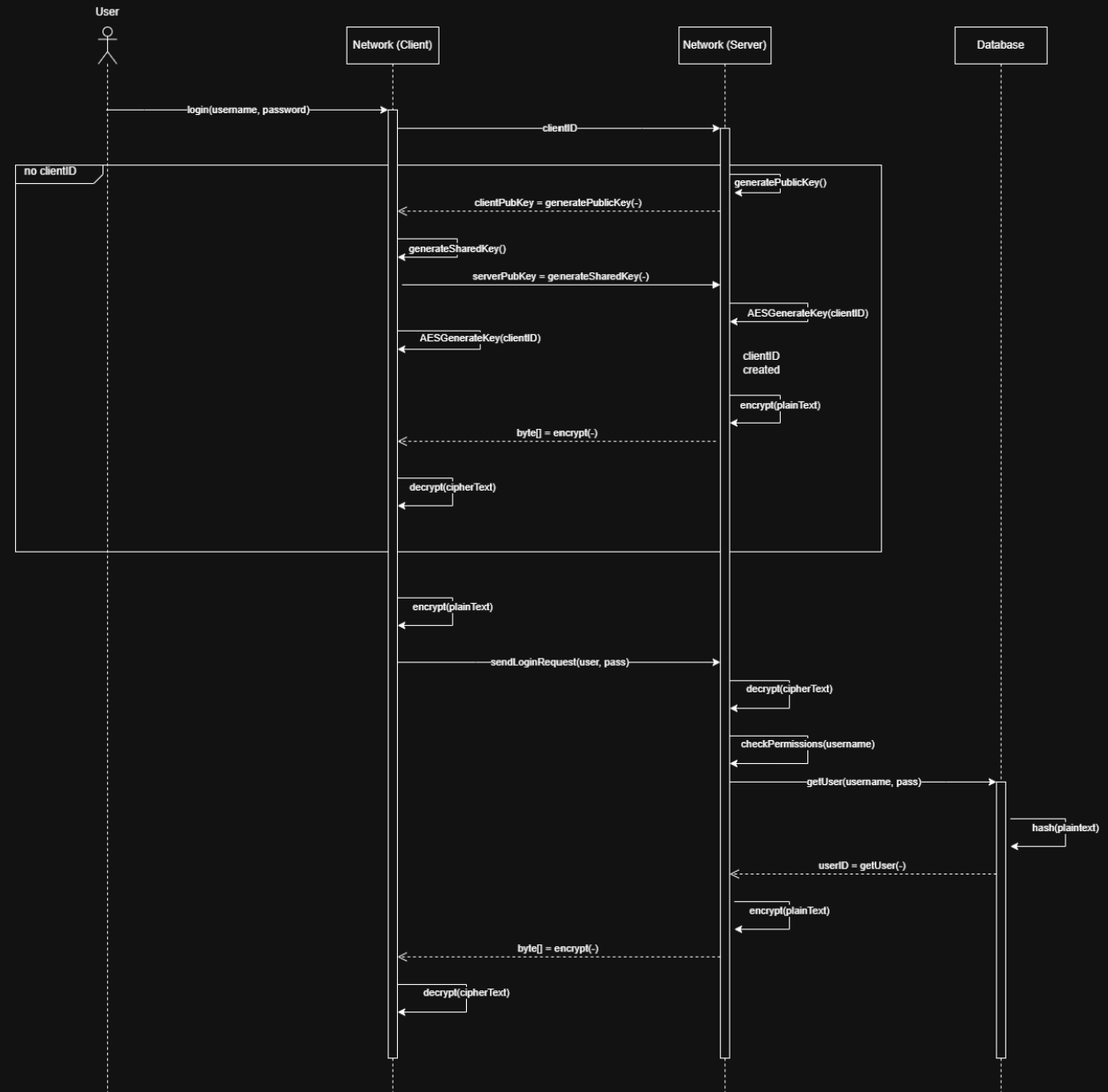
Structure Diagrams



State Diagram



Sequence Diagram



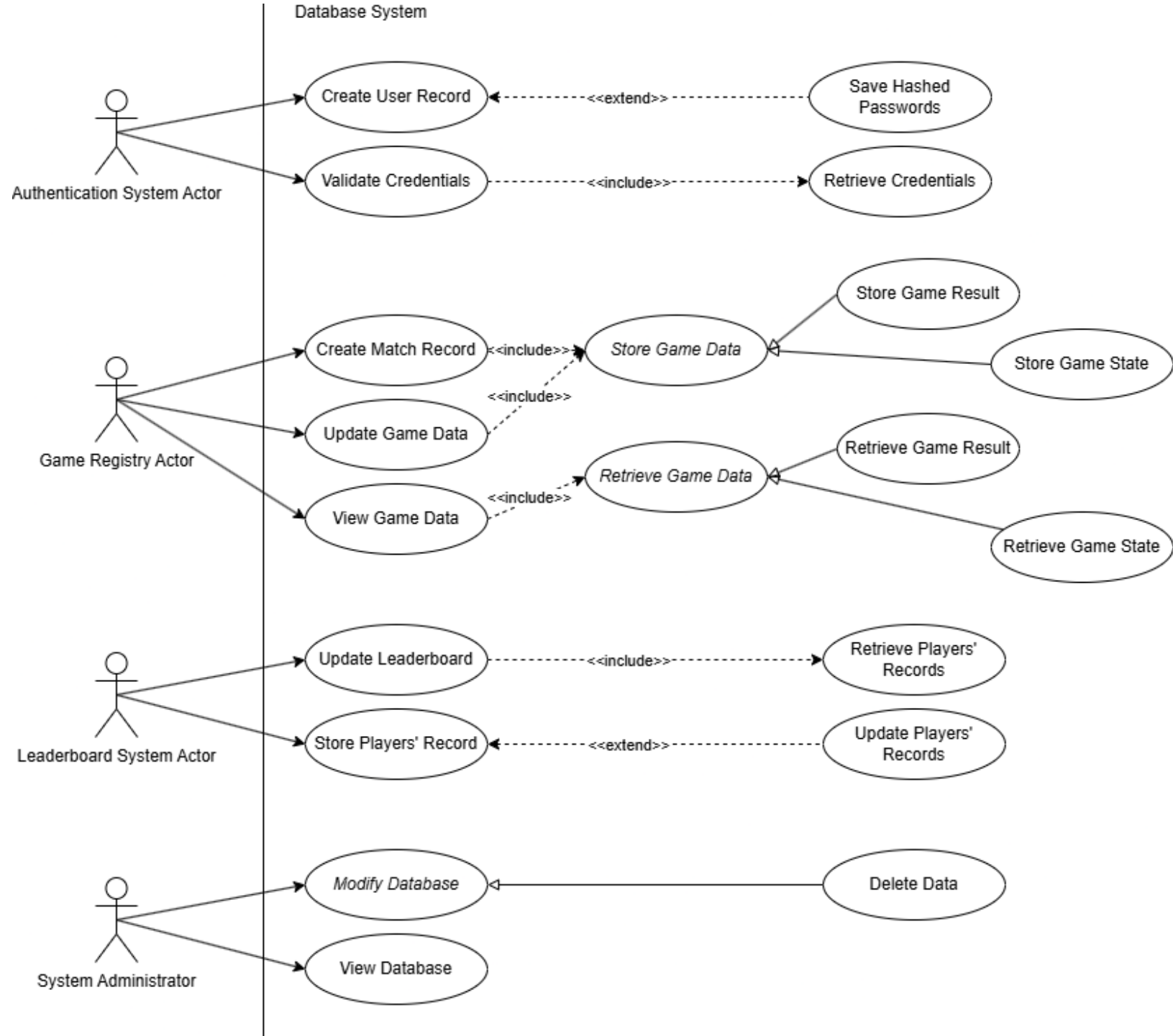
Use Case Diagram

Goal:

- Show **when** and **how** to use the database

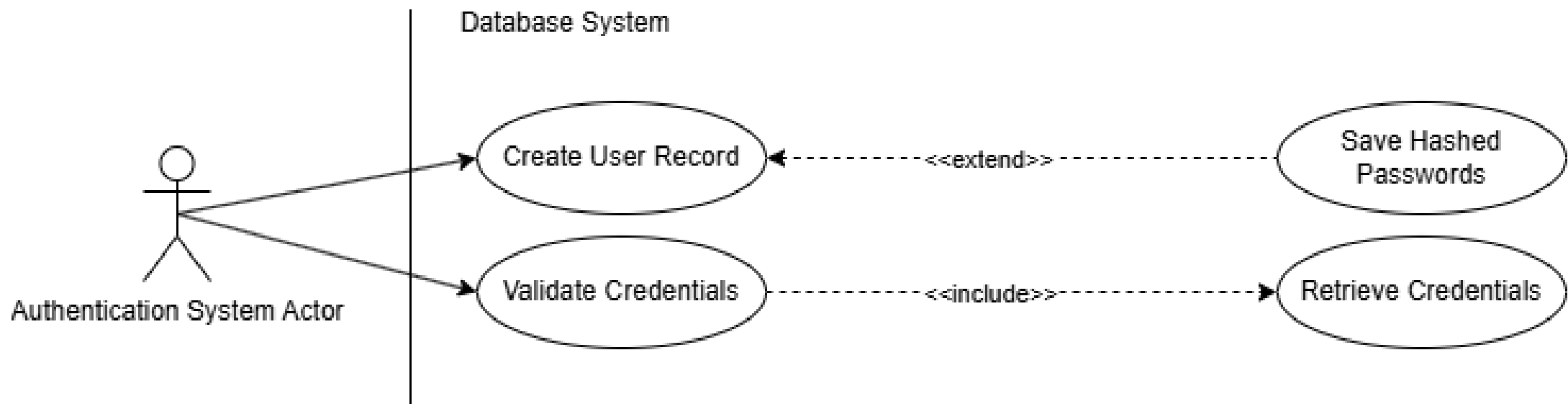
Main Actors:

- Authentication System
- Game Registry
- Leaderboard
- System Admin



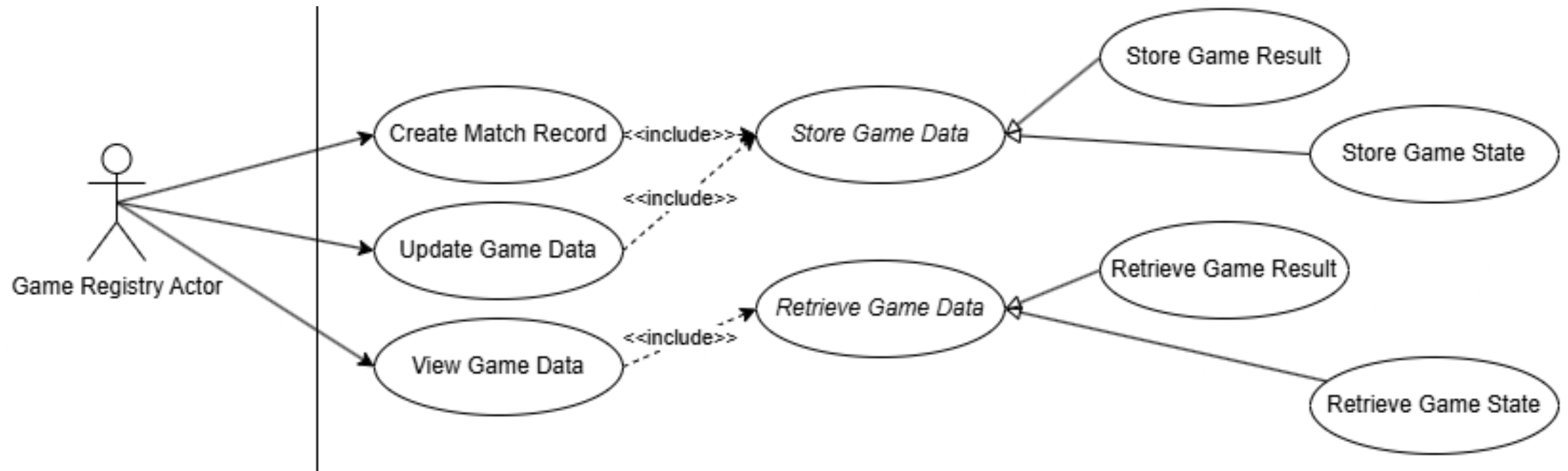
Authentication System

- **Role:** responsible for handling user login and account creation
- **Purpose:** store and confirm the user identity
- **Actions:**
 - Create user record
 - Validate credentials



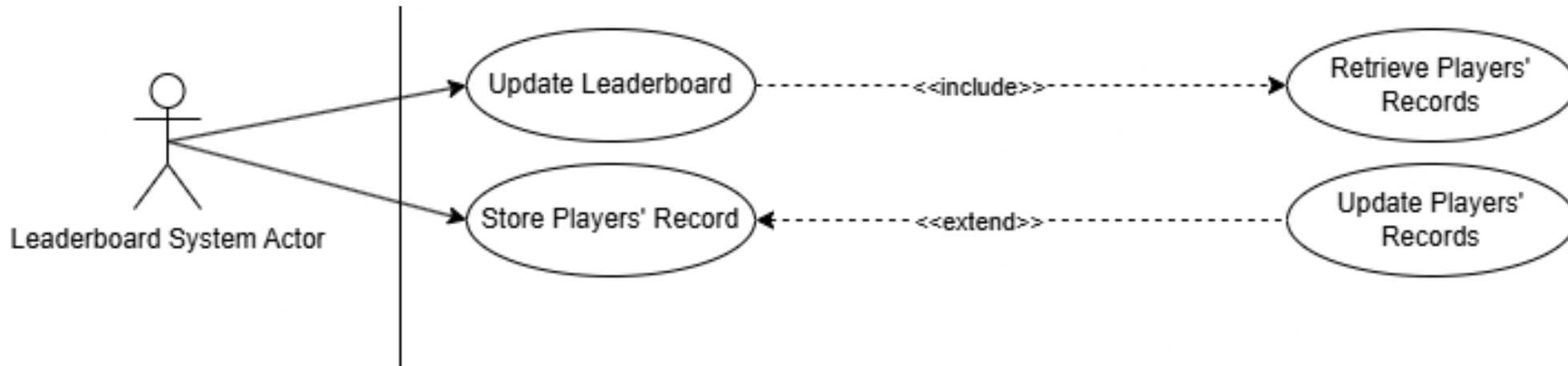
Game Registry

- **Role:** manages all match-related data
- **Purpose:** create matches and store game-related information
- **Actions:**
 - Create match record
 - Update game data
 - View game data



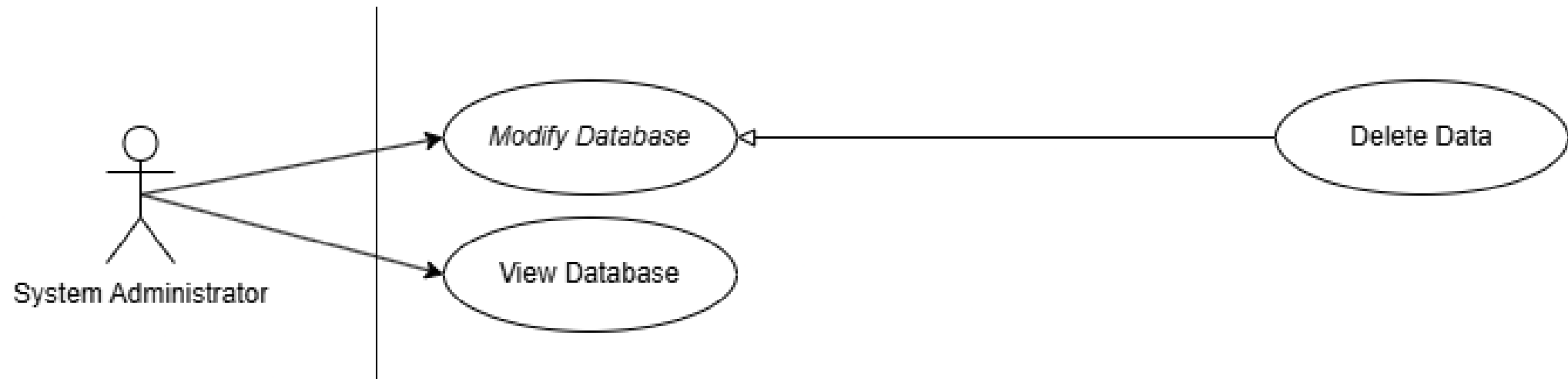
Leaderboard System

- **Role:** responsible for ranking players
- **Purpose:** update and maintain the leaderboard
- **Actions:**
 - Update leaderboard
 - Store players' record



System Administration

- **Role:** a human role that organizes the database
- **Purpose:** manage the database
- **Actions:**
 - Modify database
 - View database



Our Plan Moving Forward

As we transition from the planning to the development stage of the project, we plan on:

- Developing all 4 Core systems of the program as outlined in the diagrams. Including the Database, Game Registry, User Authentication, and Leaderboard Systems.
- Clearly articulating how and when we will be implementing different parts of the program to other teams to ensure they are not blocked
- Continue to maintain and update the planning diagrams as changes are made so that everyone stays in the loop
- Keep an active worklog on the issue boards so that the whole team is aware of the work being done in real time.

Team Integration

Position in Platform Ecosystem

Because the other teams will build off our work, we are ensuring we implement core systems and ensure other sub-teams can build off our foundation.

Upkeep of Shared Living Documents

Due to potential design changes, it will be our job to update the planning diagrams and share them with the rest of the group so that they can integrate their designs correctly.

Coordination with Integration and Release Team

Especially at the start of the implementation stage, we will have to coordinate heavily with the integration and release team to ensure that the foundation of each 4 main components of the system can merge successfully as merge errors may occur.

Potential Challenges and Issues

- **Server can only handle 1 client at a time**
- **Client code has to be manually updated to reflect server IP**
- **Client(s) and server must be on same network**

